

OSSP:

Commands:

uname: Prints system and kernel information.

lscpu: Displays the processor configuration and specifications.

```
saivihamsh@Varshini-PC:~$ gcc task1.c -o task1
saivihamsh@Varshini-PC:~$ ./task1
Enter a Linux command: uname

Parent Process
Parent PID : 440
Child PID  : 441

Child Process
Child PID : 441
Parent PID: 440

Executing command: uname
```

ps: Lists all active processes currently running in the system.

```
saivihamsh@Varshini-PC:~$ ./task1
Enter a Linux command: ps

Parent Process

Parent PID : 444
Child Process
Child PID  : 445
Child PID  : 445
Parent PID: 444

Executing command: ps

  PID TTY          TIME CMD
  306 pts/0        00:00:00 bash
  444 pts/0        00:00:00 task1
  445 pts/0        00:00:00 ps

Child process completed.
```

Relation Between Hardware Resources and Operating System Services:

1. CPU

The operating system controls how programs use the processor by executing them one after another very quickly.

Example: Running a calculator while downloading a file.

2. Memory (RAM) -> Memory Management

The OS reserves memory for each application and releases it after the program is closed.

Example: Opening Paint uses RAM, and closing it frees the memory.

3. Storage (SSD) → File System Management

The operating system manages the reading and writing of data on storage devices.

Example: Copying a file from one folder to another.

4. I/O Devices → Device Management

The OS manages communication between the computer and connected devices using drivers.